BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 8485 / 8488

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8485

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Which acid can be used as a catalyst in Friedel-Crafts reactions?

(A) HNO_3 (B) $BeCl_2$

(C) $AlCl_3$ (D) H_2SO_4

Q2. The presence of a double bond in a compound is the sign of:

(A) Saturation (B) Unsaturation

(C) Substitution (D) Elimination

Q3. Which set of hybrid orbitals has planar triangular shape:

(A) sp^2 (B) sp^3

(C) sp (D) dsp^2

Q4. The colour of transition metal complexes is due to:

(A) Paramagnetic nature of element (B) Ionization

(C) Loss of S-electron (D) d-d transition of electrons

Q5. Bleaching powder may be produced by passing chlorine over:

(A) Calcium carbonate
(B) Hydrated calcium sulphate
(C) Calcium hydroxide
(D) Magnesium hydroxide

Q6. Nitric oxide forms a brown coloured addition compound with $FeSO_4$. This test is used to confirm the presence of:

(A) Carbonates
(B) Phosphates
(C) Nitrates
(D) Sulphates

Q7. Which element forms an ion with charge +3?

(A) Beryllium
(B) Aluminum
(C) Carbon
(D) Silicon

Q8. Which ion will have the maximum value of heat of hydration?

(A) Na^{+}
(B) Cs^{+1}
(C) Ba^{+2}
(D) Mg^{+2}

Q9. Sodium reacts with excess of oxygen and forms:

(A) Sub oxide
(B) Normal oxide
(C) Peroxide
(D) Super oxide

Q10. The pH range of the acid rain is:

(A) 7 – 6.5
(B) 6.5 – 6
(C) 6 – 5.6
(D) Less than 5

Q11. In purification of potable water the coagulant used is:

(A) Nickel sulphate
(B) Copper sulphate
(C) Barium sulphate
(D) Alum

Q12. Phosphorous helps in the growth of:

(A) Root
(B) Leaves
(C) Stem
(D) Seed

Q13. Which polymer is an addition polymer?

(A) Nylon-6,6
(B) Polystyrene
(C) Terylene
(D) Epoxy resin

Q14. Which acid is used in the manufacturing of synthetic fiber?

(A) Formic acid
(B) Oxalic acid
(C) Carbonic acid
(D) Acetic acid

Q15. Which will have the highest boiling point?

(A) Mathanal (B) Ethanal

(C) Propanal (D) 2-Hexanone

Q16. Which compound shows hydrogen bonding?

(A) C_2H_5OH (B) C_2H_4

(C) C_2H_5Cl (D) CH_3-O-CH_3

Q17. For which mechanism, the first step involved is the same:

(A) E1 and E2 (B) E2 and S_n2

(C) S_n1 and E2 (D) E1 and S_n1

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8485

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Give two chemical reactions of ZnO which prove its amphoteric character.

(ii) Alkali metals give ionic hydrides. Give justification for the statement.

(iii) What happens when gypsum is strongly heated?

(iv) What is lime mortar?

(v) Why do K_2CrO_4 and $K_2Cr_2O_7$ show similar properties in aqueous solution?

(vi) Transition elements show variable oxidation state. Give reason.

(vii) What is β-elimination reaction? Give one example.

(viii) How would you prepare n-butane and ethane from ethyl chloride?

(ix) What is thermoplastic polymer? Give two examples.

(x) What is denaturation of proteins? Give one example.

(xi) Mention two points of difference between DNA and RNA.

(xii) What is requirement for a compound to be used as a fertilizer?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Define the "Ring Test" for the confirmation of the presence of nitrate ions in the solution.

(ii) Why is SO_3 dissolved in H_2SO_4 and not in water?

(iii) Write four uses of bleaching powder.

(iv) Why HF is weaker acid than HCl?

(v) Define functional group. Give the functional group of ether and carboxylic acid.

(vi) Define Tauto merism. Give an example.

(vii) How is ethyne prepared on industrial scale?

(viii) How is Raney Nickel prepared? Give its one use.

(ix) Convert ethene into ethyl alcohol.

(x) How is ozone damaged in stratosphere by chlorofluorocarbons (CFCs)?

(xi) Differentiate between primary and secondary pollutants.

(xii) How does acid rain affect environment?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What are silicates? How sodium silicate is prepared?

(ii) What is boric acid? How it is prepared in laboratory?

(iii) Why are liquid silicones preferred over ordinary organic lubricants?

(iv) Why nitration of toluene is faster than benzene?

(v) How will you distinguish between methanol and ethanol?

(vi) Describe the term esterification using ethyl alcohol as an example.

(vii) How will you distinguish between acetone and ethyl alcohol?

(viii) How would you convert acetic acid into acetamide?

(ix) What is peptide bond? Write the formula of a dipeptide.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8488

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The electrophile in aromatic sulphonation is:

(A) H_2SO_4 (B) HSO_4^-

(C) SO_3 (D) SO_3^+

Q2. When cyanogen chloride (Cl-CN) is made to react with ethyl magnesium bromide the product formed is:

(A) CH_3-CN (B) CH_3-CH_2-CN

(C) CH_3-CH_2-CH_2-CN (D) CH_2=CH-CN

Q3. Which compound will have the maximum repulsion with water?

(A) C_6H_6 (B) C_2H_5OH

(C) $CH_3CH_2CH_2$-OH (D) CH_3-O-CH_3

Q4. It will have the highest boiling point:

(A) Mathanal (B) Ethanal

(C) Propanal (D) 2-Hexanone

Q5. Which reagent is used to reduce a carboxylic group to an alcohol?

(A) H_2/Ni (B) H_2/Pt

(C) $NaBH_4$ (D) $LiAlH_4$

Q6. Nitrogeneous bases is not present in RNA:

(A) Cytosine (B) Adenine

(C) Thiamine (D) Uracil

Q7. Phosphorus helps the growth of:

(A) Root (B) Leave

(C) Stem (D) Seed

Q8. Methane has a mean residence time in atmosphere about:

(A) 3-7 years (B) 4-7 years

(C) 5-7 years (D) 6-7 years

Q9. Newspaper can be recycled again and again by how many times?

(A) 2 (B) 3

(C) 4 (D) 5

Q10. Select the two normal elements are present in seventh period:

(A) Rb , Sr (B) Cs , Ba

(C) Fr , Ra (D) La , Hf

Q11. It does not belong to alkaline earth metals:

(A) Be (B) Ra

(C) Ba (D) Rn

Q12. The chief ore of aluminum is:

(A) Na_3AlF_6 (B) $Al_2O_3 \cdot 2H_2O$

(C) Al_2O_3 (D) $Al_2O_3 \cdot H_2O$

Q13. Birkeland-Eyde process used for the preparation of:

(A) HNO_3 (B) H_2SO_4

(C) C_6H_6 (D) HCHO

Q14. The anhydride of $HClO_4$ is:

(A) ClO_3 (B) ClO_2

(C) Cl_2O_5 (D) Cl_2O_7

Q15. It is a typical transition metal:

(A) Sc (B) Y

(C) Co (D) Ra

Q16. Linear shape is associated with which set of hybrid orbitals?

(A) sp (B) sp^2

(C) sp^3 (D) dsp^2

Q17. Synthetic rubber is made by polymerization of:

(A) Chloroform (B) Acetylene

(C) Divinyl acetylene (D) Chloroprene

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1214-XII124

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) What is the difference between acidic and basic oxides? Give one example of each.

(ii) Carbon and hydrogen possess reducing properties. Show with equations.

(iii) Decomposition of lithium nitrate gives different products than the nitrates of other alkali metals. Why? Give reaction.

(iv) Give brief description of alkali and alkaline earth metal sulphates solubility in water.

(v) How is K_2CrO_4 converted into $K_2Cr_2O_7$? Show with reaction.

(vi) How is chromyl chloride obtained from potassium dichromate? Give reaction.

(vii) How is ethene prepared by E_1 elimination reaction?

(viii) Give reaction for the preparation of 1-Butanol from Grignard reagent.

(ix) What is the difference between isomerases and ligases?

(x) Give reaction for the formation of soap from triglyceride.

(xi) Give brief description of the rancidity of fats.

(xii) Give names of the nitrogeneous fertilizers.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) How does PCl_5 react with water?

(ii) Write names and formulas of any two sulphide ores of Sulphur.

(iii) Give chemical reactions of H_2SO_4 with NaCl and NaBr.

(iv) What is available chlorine?

(v) Define isomerism. Write two isomers of butane.

(vi) How was coal formed from wood under the Earth?

(vii) Write common name and structural formula of 1-methyl propene.

(viii) What is incomplete oxidation of CH_4?

(ix) Give structural formulae of the compounds: (i) Potassium maleate (ii) Disodium succinate

(x) How is mustard gas prepared from ethene?

(xi) What are primary pollutants? Give two examples.

(xii) What is leachate?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) How is H_3BO_3 prepared from (i) Borax (ii) Colemanite

(ii) Write four uses of borax.

(iii) Why are the liquid silicones preferred over the ordinary organic lubricants?

(iv) How is ethyl benzene prepared by Wurtz-Fitting reaction?

(v) How are the phenols prepared by Dow's method?

(vi) Describe the term esterification using ethyl alcohol as an example.

(vii) Define silver mirror test. Give the reaction involved.

(viii) How is acetic acid converted into (i) Methane (ii) Acetic anhydride

(ix) What is Zwitter ion? Give its structural formula.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.